GROUP PRESENTATION ALLOCATIONS

COURSE CODE: **SWE2102**

COURSE NAME: **REQUIREMENTS ENGINEERING**

YEAR: **BSE II**

DATE	GROUP	TOPIC
8th September	F	1. Introduction to Requirements Engineering 2. Overview of Software Engineering 3. Importance of software requirements 4. Role of requirements engineering in software development 5. Types of Requirements -Functional vs. Non-functional requirements -User vs. System requirements 6. Introduction to the Requirements Engineering Process (Elicitation, analysis, specification, validation, and management)
15th September	K	1. Requirements Elicitation 2. Elicitation Techniques (Interviews, questionnaires, workshops, brainstorming sessions Observations, prototyping, use cases, and scenarios) 3. Challenges in Elicitation 4. Stakeholder Analysis -Identifying and managing stakeholders -Techniques for engaging stakeholders -Common pitfalls and how to avoid them
22nd September	C	• Requirements Specification • Writing Effective Requirements • Characteristics of good requirements: clear, concise, testable • Use of templates and standards • Documenting Requirements • Software Requirements Specification (SRS) documents • IEEE standards for SRS • Use of Natural Language vs. Formal Specifications

		• Advantages and disadvantages of each approach
29th September	G	• Requirements Specification • Writing Effective Requirements • Characteristics of good requirements: clear, concise, testable • Use of templates and standards • Documenting Requirements • Software Requirements Specification (SRS) documents • IEEE standards for SRS • Use of Natural Language vs. Formal Specifications • Advantages and disadvantages of each approach
06th October	A	• Requirements Validation • Techniques for Validation • Reviews, inspections, and walkthroughs • Prototyping and simulation • Requirements Traceability • Traceability matrices • Ensuring consistency and completeness • Validation Tools • Overview of tools used in industry
13th October	I	• Requirements Management • Introduction to Requirements Management • Importance of managing changing requirements • Version control and configuration management • Change Management • Change control boards (CCB) • Impact analysis and risk management • Tools for Requirements Management • Overview of popular tools: JIRA, IBM DOORS, etc.
20th October	B	• Agile Requirements Engineering • Agile Methodologies Overview • Scrum, Kanban, and Lean principles • Agile Requirements Practices • User stories, epics, and acceptance criteria • Backlog grooming and sprint planning • Role of the Product Owner • Managing the product backlog • Communication with stakeholders

20th October	H	• Requirements for Non-functional Aspects • Non-functional Requirements (NFRs) • Categories: performance, security, usability, etc. • Elicitation and documentation of NFRs • Quality Attributes • Measuring and specifying quality attributes • Trade-offs between different attributes
27th October	E	• Case Studies and Best Practices • Case Study Analysis • Real-world examples of requirements engineering • Lessons learned from successful and failed projects • Industry Best Practices • Guidelines for effective requirements engineering • Common pitfalls and how to avoid them
03rd November	D	• Requirements Engineering Tools and Techniques • Introduction to RE Tools • Overview of popular tools: Rational RequisitePro, CaliberRM, etc. • Hands-on Session • Using a selected tool for requirements management • Tool Evaluation and Selection • Criteria for selecting tools for a project
10th November	J	• Requirements in the Software Development Lifecycle • Integration with Software Development Models • Waterfall, V-Model, Spiral, and Agile • Role of Requirements in Different Phases • Design, development, testing, and maintenance • Managing Requirements in Large Projects • Scaling techniques and approaches